Dr. Steven R. Goldman

Senior Science Software Engineer · Space Telescope Science Institute
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Specialization: Image alignment and resampling, evolved stars, dust, long period variables, symbiotic Miras, Magellanic Clouds, dwarf galaxies, Local Group, M31, UV to mid-IR imaging and spectroscopy, radio interferometry, spectral energy distribution (SED) fitting, and time-series analysis.

Research Interests: The impact of evolved stars on their environment and vice-versa. Specifically, the effects of metallicity on the dust production, wind dynamics, mass-loss mechanism, and evolution of Asymptotic Giant Branch (AGB) stars and Red Supergiants (RSGs). This involves statistical samples of nearby stellar populations targeted with the Hubble, Spitzer, and James Webb Space Telescopes.

APPOINTMENTS

Space Telescope Science Institute (STScI) February 2023 – present
Baltimore, MD USA
Position: Science Software Engineer − Hubble Space Telescope Calibration Pipeline Lead

Haverford College January 2023 – May 2023
Haverford, PA USA
Position: Visiting Assistant Professor

Stratospheric Observatory For Infrared Astronomy (SOFIA) November 2021 – February 2023
Mountain View, CA USA
Position: Observatory & Instrument Scientist (Research Scientist-Track)

STScI October 2017 – November 2021
Baltimore, MD USA
Position: Postdoctoral Fellow
Supervisor: Martha Boyer

EDUCATION

Keele University October 2013 – December 2017
Degree: PhD in Astrophysics
Thesis: The metallicity dependence of maser emission and mass loss from RSG and AGB stars
Advisor: Jacco Th. van Loon

St. Lawrence University September 2009 – July 2013
Degree: B. S. in Physics

AWARDED PROPOSALS (PRINCIPAL INVESTIGATOR)

Total PI funding secured: $480 k

2025 Hubble Space Telescope, PID: 18148 ($29 k)	archival
2025 James Webb Space Telescope, PID: 6852 ($209 k)	30.2 hours
2022 Hubble Space Telescope, PID: 17088 ($137 k)	30 orbits
2021 Hubble Space Telescope, PID: 16492 ($65 k)	4 orbits
2021 SOFIA, PID: 75_0057 ($40 k)	4 hours
2017 Australia Telescope Compact Array Telescope, PID: C2996	92 hours

2017 Very Large Telescope VISIR, PID: 099.D-0907	1 Night
2017 Very Large Telescope VISIR, PID: 098.D-0272	0.5 hours
2016 Very Large Telescope XSHOOTER, PID: 097.D-0605	1.5 hours
2015 Westerbork Synthesis Radio Telescope, PID: R14/010	30 hours
2014 Southern African Large Telescope	5.5 hours
2012 National Science Foundation Summer REU Fellowship	

RECENT AWARDED PROPOSALS (CO-INVESTIGATOR)

James Webb Space Telescope (5 programs)	113 primary / 2 parallel hours
Hubble Space Telescope (8 programs)	107 primary / 520 parallel orbits
VLT/VLTI (2 programs)	19.3 hours
SOFIA (2 programs)	20.25 hours
ACA (NESS)	750 hours
Astrophysics Data Analysis Program (18-ADAP18-142)	$335 k
ALMA	5 hours

SELECTED SEMINARS & CONFERENCE PRESENTATIONS

2026 Talk	Ursinus College, USA
2025 Talk, Why galaxies care about AGB stars V	Santiago, Chile
2024 Talk, AAS Winter Meeting	New Orleans, USA
2022 Invited Speaker, IAU General Assembly	Busan, South Korea
2022 Featured Speaker, AstroPhilly22	Villanova University, USA
2022 Invited Colloquium	University of Auckland, New Zealand
2022 Invited Colloquium	Auckland University of Technology, New Zealand
2022 Invited Colloquium	University of Canterbury, New Zealand
2022 Invited Talk, AAS Summer Meeting	Pasadena, USA
2022 Invited Colloquium	NRC Herzberg, Canada
2021 Invited Panelist, Evolved Stars and their Circumstellar Environments	Global (Remote)
2021 Invited Talk, GAPS: unsolved problems in red Giants And suPergiantS	Global (Remote)
2021 Talk, DELVE: The Death-Throes of Evolved Stars	Global (Remote)
2021 Invited Talk	St. Lawrence University, USA
2020 Invited Talk	STScI, USA
2019 Invited Colloquium	RIT, USA
2019 Invited Colloquium	EAO Hawaii, USA
2019 Talk, A Star Has Evolved: A Conference in the Honor of Hans Olofsson	Smögen, Sweden
2019 Talk, HotSci	STScI, USA
2018 Talk, Cosmic Dust: origin, applications & implications	Copenhagen, Denmark
2018 Talk, European Week in Astronomy and Space Science	Liverpool, UK
2016 Talk, Blowing in the wind (Awarded Best Talk)	ICISE, Vietnam
2016 Talk, Postgraduate Research Symposium (Awarded Best Talk)	Keele University, UK
2016 Talk, SKA Delivering the Science	Cambridge University, UK
2015 Talk, UK SKA Science Meeting	Manchester University, UK
2015 Invited Colloquium	Kagoshima University, Japan

TEACHING

Haverford College, USA
Astro 101, Visiting Assistant Professor — Spring, 2023

Keele University, UK
Electronics, Demonstrator/Lab Assistant — 2014 – 2016

Programming I, Demonstrator/Lab Assistant 2015

STUDENTS MENTORED

Ava Allen STScI, Postbac 2026
Tracing the dynamical history of DDO 68 with evolved stars

Tessa Pearlstein Haverford College, Summer 2023
Dust Injection Rate of TP-AGB stars in the M33 galaxy

Sean Garner SOFIA/USRA, Summer 2022
SOFIA/EXES Water Line Analysis of the Symbiotic Mira HM Sge

Nathan Wolthuis SOFIA/USRA, Summer 2022
*HST/WFC3 Image Analysis of Nebular N*II *emission of HM Sge*

OBSERVING EXPERIENCE

Hubble Space Telescope	34 orbits
James Webb Space Telescope	30 hours
SOFIA	3 flights
James Clerk Maxwell Telescope	70 hours
Very Large Telescope	1 night
Australia Telescope Compact Array	92 hours
Parkes Radio Telescope	36 hours
Arecibo L-band (ALFALFA)	20 hours

SOFTWARE DEVELOPMENT

Observatory Service

- *Drizzlepac*: Primary maintainer for the widely-used image re-alignment software also used for the alignment, drizzling, and catalog creation for the Hubble Advanced Products.
- *SOFIA Data-Analysis Cookbooks and FAQs*: Jupyter notebooks demonstrating a variety of techniques for analyzing infrared data from SOFIA, as well as detailed explanations (FAQs) of common misconceptions and mistakes.

Research

- The *DESK*: Independently led the development and publication of the first open-source python package to fit the SEDs of evolved stars with models to get meaningful intrinsic properties.
- The *BEAST*: Improved on testing and functionality of a large-scale open-source astropy-affiliated python package that fits stellar populations' SEDs to determine ISM, dust, and stellar properties using Bayesian statistics, dust models, and stellar evolutionary models.

COMMUNITY OUTREACH

Public Outreach Talks, 2018 – Present, (USA, Canada, & Mexico)
- NASA's *Universe of Learning*, *Skype-a-Scientist*, *Astronomy On Tap*, Independently organized

Earth and Space Observatory volunteer, 2013 – 2017, (Keele University, UK)

ACADEMIC SERVICE

STScI Spring Symposium LOC	May 2025
Organized the SOFIA New Zealand Lecture Series	July 2022
SOFIA School Co-Chair	February 2022
STScI postdoctoral representative	2020 – 2022
DELVE conference SOC	February 2021
Low-Density Universe Lunch Organizer	2018 – 2019

TAC Panel Support: Hubble & SOFIA
Panelist: NASA ADAP
Referee: Astrophysical Journal; Astronomy & Astrophysics; Journal of Open Source Software

SELECTED PRESS

CNN, NYT, Guardian, *Astronomers take the first close-up picture of a star outside our galaxy* (Link)
NASA, *NASA's Hubble Finds Surprises Around a Star That Erupted 40 Years* (Link)
Phys.org, *Research investigates evolution of symbiotic binary HM Sagittae* (Link)
Sky at Night magazine interview about SOFIA New Zealand deployment (Link)
NASA, *SOFIA Watches a Binary Star System's Eclipse* (Link)

REFEREED PUBLICATIONS

Links: NASA ADS, ORCID, Google Scholar

Summary: I have published 37 refereed journal articles, with >574 total refereed citations, and an h-index of 13. Papers with >15, >25, and >150 citations are marked with 1, 2, or 3 asterisks, respectively.

First-Authored Peer-Reviewed Publications:

1. Goldman S. R. et al., 2025, ApJ, 980, 191
   Equatorial Enhancement in the Dustiest OH/IR Stars in the Galactic Bulge
2. Goldman S. R. et al., 2024, ApJ, 961, 14
   A Multi-Wavelength Study of the Symbiotic Mira HM Sge with SOFIA & HST
3. [*] Goldman S. R. et al., 2022, ApJS, 259, 41
   A Census of Thermally-Pulsing AGB stars in the Andromeda Galaxy and a First Estimate of their Contribution to the Global Dust Budget
4. Goldman S. R., 2020, JOSS, 5(54), 2554
   The Dusty Evolved Star Kit (DESK): A Python package for fitting the Spectral Energy Distribution of Evolved Stars. Journal of Open Source Software
5. Goldman S. R. et al., 2019, ApJ, 884, 152
   AGB Stars in the Nearby Galaxy: Leo P
6. [**] Goldman S. R. et al., 2019, ApJ, 877, 49
   An Infrared Census Of Dust In Nearby Galaxies With Spitzer (DUSTiNGS): V. The Period-luminosity Relation For Dusty Metal-poor AGB Stars
7. [*] Goldman S. R. et al., 2018, MNRAS, 473, 3835
   A dearth of OH/IR stars in the Small Magellanic Cloud

***8. Goldman S. R. et al., 2017, MNRAS, 465, 403
The wind speeds, dust content, and mass-loss rates of evolved AGB and RSG stars at varying metallicity

Co-Authored Peer-Reviewed Publications:

9. Groenewegen et al. 2026, submitted to A&A
CO rotational line emission in very red carbon stars in the Magellanic Clouds

10. Gull et al. 2026, submitted to ApJ
Unveiling Massive Main-Sequence Stars in Sextans A through Panchromatic Photometry

11. Lindberg et al. 2026b, submitted to ApJ letters
Scylla VII: Observational Evidence for an Order of Magnitude in Dust Mass Opacity Evolution with ISM Density in the Large Magellanic Cloud

12. Gavetti et al. 2026, accepted in ApJ
Dust in the very metal-poor galaxy Sextans A with JWST. I: Characterizing the evolved stellar population of Sextans A based on JWST observations and stellar evolution models

13. Lindberg et al. 2026a, accepted in ApJ
Scylla VI: Parsec-Scale Dust Extinction Maps in the SMC and LMC

14. Wallström et al. 2025, A&A, 704, 276
The Nearby Evolved Stars Survey: III. First data release of JCMT CO-line observations

15. Boyer et al. 2025, ApJ, 991, 24
SiC and metallic iron dust around asymptotic giant branch stars at primordial metallicity in the Sextans A dwarf galaxy

16. Gull et al. 2025a, ApJ, 986, 25
A Low Metallicity Massive Contact Binary Star System Candidate in WLM identified by Hubble and James Webb Space Telescope imaging

17. Omelian et al. 2025, ApJ, 984, 128
SOFIA/FORCAST/HAWC+ Monitoring of the Dust Emission from R Aqr: Eclipse and Periastron Passage

18. Lindberg et al. 2024, ApJS, 276, 8
Scylla IV: Characterizing Resolved Stars and Line of Sight Extinction in the Magellanic Clouds with Bayesian SED Fitting Techniques

19. Ohnaka et al. 2024, A&A, 691, 9
Imaging the innermost circumstellar environment of the red supergiant WOH G64 in the Large Magellanic Cloud

**20. McQuinn et al. 2024, ApJ, 976, 60
The Ancient Star Formation History of the Extremely Low-Mass Galaxy Leo P: An Emerging Trend of a Post-Reionization Pause in Star Formation

21. Gilbert et al. 2024 , ApJS, 276, 8
The Local Ultraviolet to Infrared Treasury I. Survey Overview of the Broadband Imaging

22. Cohen et al. 2024b, ApJ, 975, 43
Scylla III. The Outside-In Radial Age Gradient in the Small Magellanic Cloud and the Star Formation Histories of the Main Body, Wing and Outer Regions

23. Cohen et al. 2024a, ApJ, 975, 42
Scylla II. The Spatially Resolved Star Formation History of the Large Magellanic Cloud Reveals an Inverted Radial Age Gradient

24. Murray et al. 2024, ApJS, 275, 5
Scylla I: A pure-parallel, multi-wavelength imaging survey of the ULLYSES fields in the LMC and SMC

25. Karnath et al. 2024, ApJ, 974, 65
An Analysis of the Complex and Compact Outflow Cavity Carved by HOPS 361-A in NGC 2071 IR

*26. Gull et al. 2022, ApJ, 941, 206
A Panchromatic Study of Massive Stars in the Extremely Metal-Poor Local Group Dwarf Galaxy Leo A

27. Ingallinera et al. 2022, MNRAS, 512, 21
Evolutionary map of the Universe (EMU): 18-cm OH-maser discovery in ASKAP continuum images of the SCORPIO field

28. Sankrit et al. 2022, ApJ, 926, 177
SOFIA/FORCAST Monitoring of the Dust Emission from R Aqr: Start of the Eclipse

**29. Scicluna et al. 2022, MNRAS, 512, 1091
The Nearby Evolved Stars Survey II: Constructing a volume-limited sample and first results from the James Clerk Maxwell Telescope

30. Jones et al., 2021, MNRAS, 504, 565
Infrared variable stars in the compact elliptical galaxy M32

31. Girardi et al., 2020, ApJ, 901, 19
PHAT XX. AGB stars and other cool giants in M31 star clusters

**32. Nanni A. et al., 2019, MNRAS, 487, 502
The mass-loss, expansion velocities and dust production rates of carbon stars in the Magellanic Clouds

33. Dharmawardena T. E. et al., 2019, MNRAS, 489, 3218
The Nearby Evolved Stars Survey: I. JCMT/SCUBA-2 Sub-millimetre detection of the detached shell of U Antliae

**34. Karambelkar V. R. et al., 2019, ApJ, 877, 110
SPIRITS Catalog of Infrared Variables: Identification of Extremely Luminous Long Period Variables

*35. Orosz G. et al., 2017, AJ, 153, 1190
Astrometry of OH/IR stars using 1612 MHz hydroxyl masers. I. Annual parallaxes of WX Psc and OH138.0+7.2

**36. Groenewegen M. A. T. et al., 2016, A&A, 596, A50
The ALMA detection of CO rotational line emission in AGB stars in the Large Magellanic Cloud

*37. McDonald I. et al., 2015, MNRAS, 453, 4324
ALMA reveals sunburn: CO dissociation around AGB stars in the globular cluster 47 Tucanae

Other Publications:

38. Goldman et al. 2022, Research Notes of the AAS, 6, 159
Sudden Dimming of the Symbiotic Mira HM Sge

39. Goldman et al. 2022, AAS Meeting, 240 proceedings, 426.04
A Census of Thermally-Pulsing AGB stars in the Andromeda Galaxy and a First Estimate of their Contribution to the Global Dust Budget

40. Sloan et al. 2021, AAS Meeting, 237 proceedings, 541.16
Spitzer's Last Look at the Small Magellanic Cloud.

41. Scicluna et al. 2020, arXiv:2002.03100 (Decadal white paper)
Studies of Evolved Stars in the Next Decade: EAO Submillimetre Futures White Paper Series

42. Karambelkar et al. 2020, , AAS Meeting, 235 proceedings, 335.04
SPIRITS catalog of infrared variables : Identification of extremely luminous long period variables

43. Goldman S. R. and Boyer M. L., 2019, IAU Meeting 343 proceedings, 14(S343), 406-408
Infrared light curves of dusty & metal-poor AGB stars

44. Goldman S. R. and Boyer M. L., 2019, AAS Meeting 233 proceedings, 33.06
Infrared light curves of dusty & metal-poor AGB stars